

Physics lab 4

August 17, 2026

Q1: Interference

Method: We simulated the time-dependent Schrödinger equation for a Gaussian wave packet reflecting off an infinite potential wall. The simulation was run at two different mesh resolutions ($N = 2^{13}$ and $N = 2^{14}$) to observe the resulting probability density as the incident and reflected waves overlap.

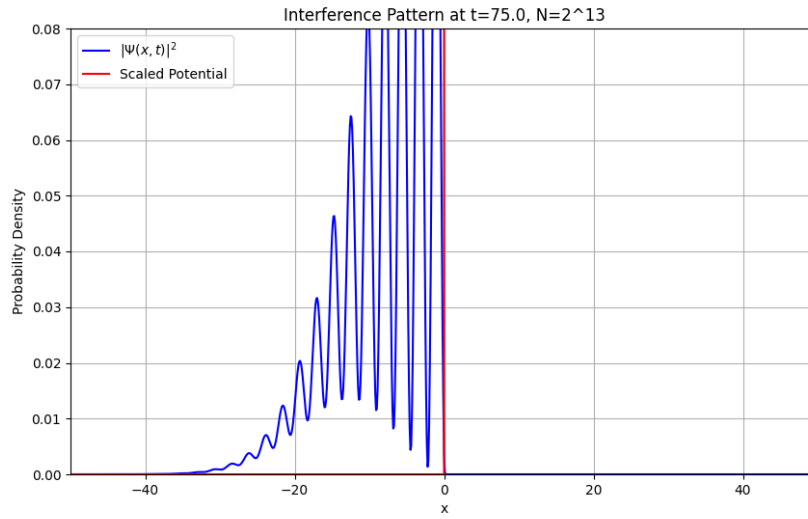


Figure 1: Interference Pattern ($N = 2^{13}$)

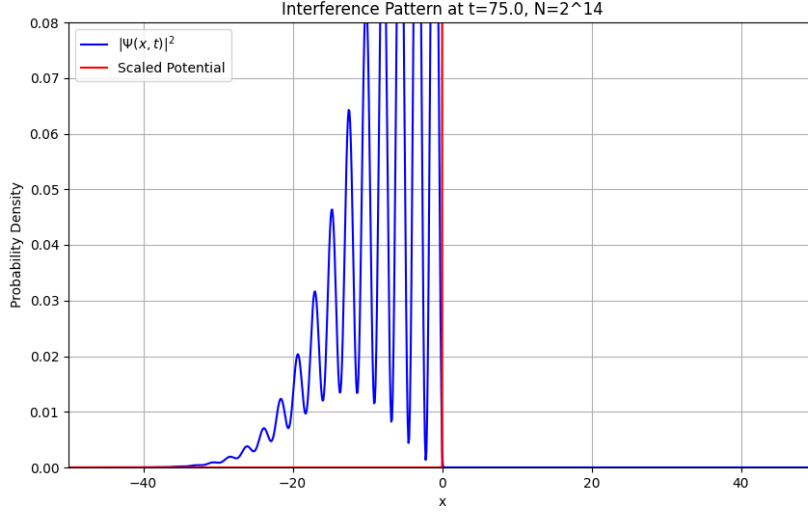


Figure 2: Interference Pattern ($N = 2^{14}$)

Theory: According to quantum mechanical theory, we expect the incident and reflected wave packets to superimpose, creating an interference pattern with spatial oscillations in the probability density $|\Psi|^2$. The physical interference pattern should converge as the numerical resolution is increased.

Results: The plots show a distinct interference pattern during the reflection. The pattern remains consistent across both mesh resolutions ($N = 2^{13}$ and $N = 2^{14}$), providing evidence that the observed interference pattern is not strongly dependent on the numerical grid resolution.

Q2: Gamow Tunneling

Method: We simulated the splitting of a wave packet colliding with the Gamow Coulomb barrier $V = \begin{cases} -V_0 & \text{for } x < R \\ C/x & \text{for } x > R \end{cases}$ where $C = R = 1$ and $V_0 = 0$ with $E = 0.9$. We then calculated the transmission probability T for several initial energies E and plotted $\ln T$ against $1/\sqrt{E}$.

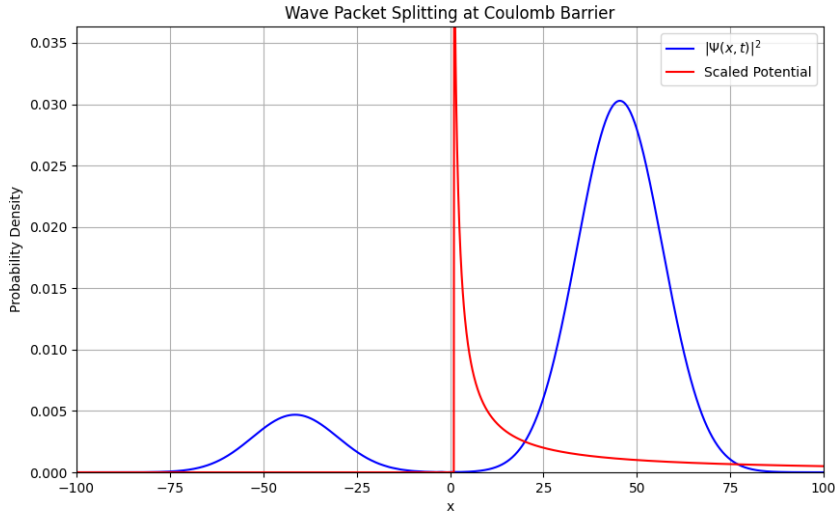


Figure 3: Wave Packet Splitting at Coulomb Barrier, $E = 0.9$

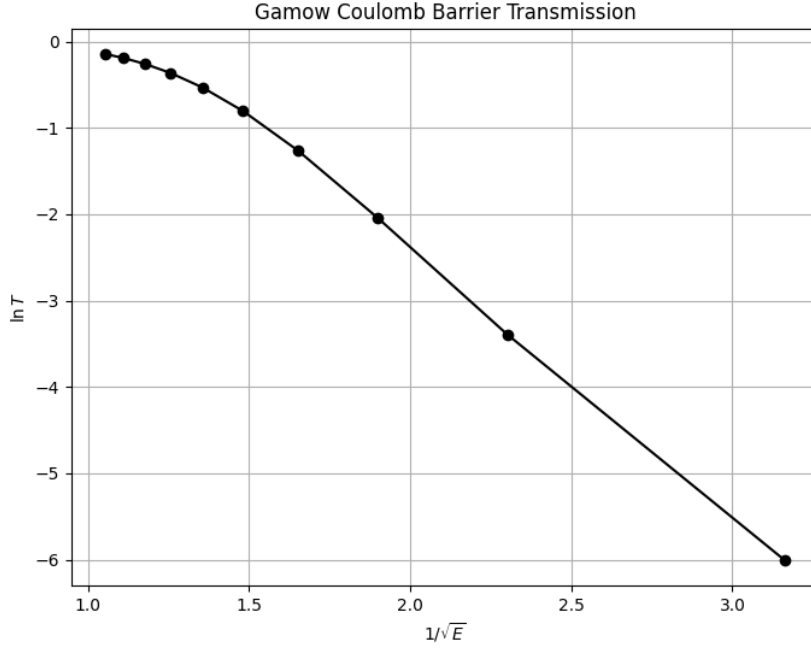


Figure 4: Transmission Probability T vs kinetic energy $E \in [0.1, 0.9]$

Theory: According to Gamow's theory of alpha decay, quantum tunneling through a Coulomb barrier predicts a transmission probability that depends exponentially on energy. Specifically, the theory states that $T \propto \exp(-k/\sqrt{E})$, which implies that a plot of $\ln T$ versus $1/\sqrt{E}$ should yield a straight line with a negative slope. Consequently, from the approximation $T = \frac{Y_T}{Y_T + Y_R}$, at lower energies we expect T to be low, which requires Y_T to be much smaller than Y_R (i.e. the transmitted wave packet is smaller than the reflected one).

Results: In fig 3, we observe that the wave packet splits into a reflected and transmitted component when encountering the Coulomb barrier, consistent with the tunneling effect predicted by quantum mechanics. We also observe that the transmitted wave packet is smaller in magnitude than the reflected one, consistent with the low transmission probability at this energy.

In fig 4, the plot of $\ln T$ against $1/\sqrt{E}$ shows an approximately linear relationship at lower energies. This approximately linear region is consistent with Gamow's predicted exponential dependence.

Q3: Fowler-Nordheim Tunneling

Method: We initialized a Gaussian wave packet with width $a = 10.0$, centered at $x_0 = -100.0$, and an initial energy $E = 0.5$. The simulation utilized a spatial grid of $N = 2^{14}$ points with a step size $dx = 0.1$ and a time step $dt = 0.01$. We set the reduced Planck constant $\hbar = 1.0$ and the particle mass $m = 1.0$.

First, we simulated the wave packet encountering a flat potential step $V(x) = V_0 = 1.0$ for $x > 0$ to observe total reflection, since $E < V_0$. Then, we simulated field emission tunneling through a slanted potential barrier $V(x) = V_0(1 - x/L)$ for $0 < x < L$ with $L = 4.0$.

We then calculated the transmission T past $x = 4.0$ for varying barrier heights $V_0 \in [0.6, 1.0]$ and plotted $\ln T$ against $V_0^{3/2}$.

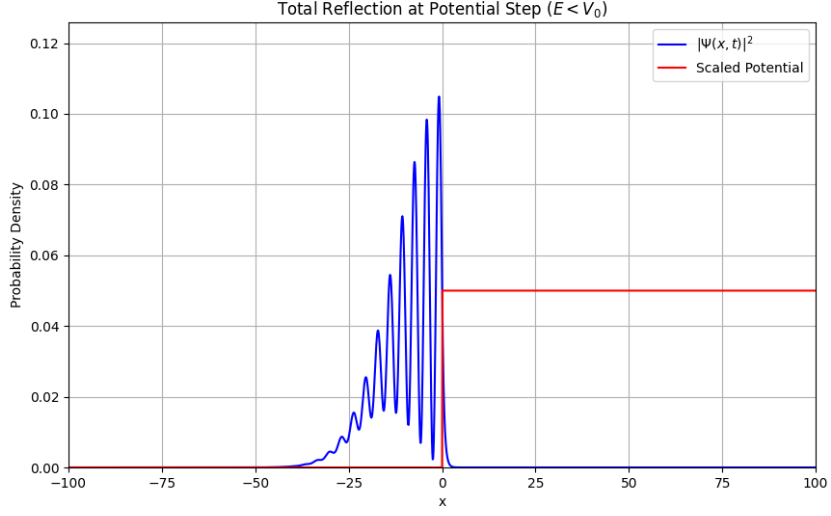


Figure 5: Complete Reflection at Flat Potential Step, $V_0 = 1.0$, $L = 4.0$, $E = 0.5$

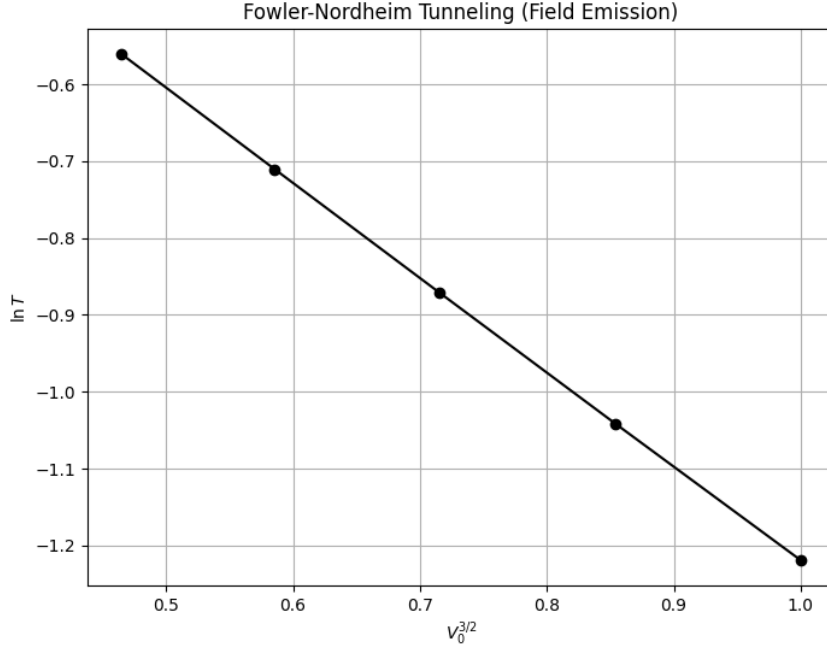


Figure 6: Fowler-Nordheim Tunneling, $\ln T$ vs $V_0^{3/2}$ for $V_0 \in [0.6, 1.0]$

Theory: According to the Fowler-Nordheim theory of field emission, the transmission probability through a triangular potential barrier is given by $T \propto \exp(-c \cdot V_0^{3/2}/F)$. Therefore, we expect a plot of $\ln T$ versus $V_0^{3/2}$ to yield a straight line.

Results: Figure 5 shows negligible transmission and predominantly reflected probability density, as expected for a potential step with $E < V_0$. Figure 6, the plot of $\ln T$ against $V_0^{3/2}$ for the slanted potential, shows an approximately linear relationship, which is consistent with the exponential dependence predicted by Fowler-Nordheim theory.